Research Statement

Sanghyun Han

My research examines how public policy shapes the gains from trade and technological change, and how those gains are distributed across countries and regions. I work at the intersection of international trade, agricultural economics, and spatial economics. My projects measure the value of international trade agreements, the consequences of regulating new technologies, and the contribution of transportation infrastructure. These topics are closely connected: policy choices and transportation networks shape producers' access to markets and opportunities to adopt innovations.

I apply structural economic models to examine how economies would respond to alternative policies or disruptions. I combine economic theory with trade and tariff data, market prices, and evidence from crop trials. The models account for how households, firms, and governments adjust their decisions, allowing me to quantify aggregate gains and losses and their distribution.

Trade agreements and international cooperation. My job market paper, *Cooperation, Commitments, and the Value of the World Trading System*, asks how much the world gains from the rules that restrain tariffs, and how that value changes with the trading environment. I distinguish cooperation from commitments under World Trade Organization rules and preferential trade agreements. Commitments limit the tariffs a government can choose, while cooperation affects how it weighs domestic and foreign welfare. I infer the degree of cooperation within the model from positive tariffs below these ceilings, where the formal limits do not bind. These choices imply that governments value a dollar of real income abroad at about half a dollar at home.

In the modeled economy, the two restraints raise world real income by about 0.8 percent relative to a world without them. The 2018–2019 tariff conflict between the United States and China lowers world real income but raises the value of the remaining system: it makes further tariff increases on other trading partners less costly, while commitments limit this response. Projected climate change instead leaves the system's aggregate value nearly unchanged, since the largest productivity changes occur in agriculture, a small share of modeled trade. These applications show why the economic damage from a shock and the value of trade rules need not move together.

Agricultural biotechnology and climate adaptation. Genetically modified crops can raise yields, yet many countries prohibit growing or importing them. In joint work with Sandro Steinbach, Carlos Zurita, and Felipe Barbosa, I study how these restrictions affect technology adoption and who benefits from it. We combine crop trials and climate data with a model of farmers' technology choices, accounting for changes in trade and land use as adoption spreads. For example, we find that higher yields can expand farmland because greater profitability draws new land into production. This work connects technology regulation to the distribution of economic gains and to environmental outcomes.

In ongoing work with Sandro Steinbach, I examine whether agricultural biotechnology can help economies adapt to climate change. We distinguish broader access to existing traits from the development of new ones, including drought tolerance. Our preliminary results indicate that biotechnology can meaningfully mitigate the economic losses associated with climate change. We are examining how these benefits depend on the traits available and the regulations governing their adoption. This project extends my research on technology diffusion to the role of innovation and regulatory policy in climate adaptation.

Transportation infrastructure and supply chains. The gains from trade also depend on the links between producers and consumers. My work on the Mississippi River asks what the United States would lose if commercial barge navigation stopped. The model allows shippers to switch modes and routes, accounting for the adjustments that can offset the disruption. The estimated annual welfare benefit of navigation is about eight times the system's annualized federal cost. Losses are concentrated in regions that depend on the river and spread through connections between agriculture and its input suppliers. This project shows how infrastructure supports economic activity beyond the industries that use it directly.

A tariff charged at the border likewise spreads beyond its point of impact. In joint work on U.S. beef tariffs with Wuit Yi Lwin, Sandro Steinbach, Carlos Zurita, and Shawn Arita, we use three decades of price data to estimate how price changes move through the supply chain. In our baseline simulations, retailers absorb more of the burden than consumers, while cattle producers gain. The division of losses changes with the extent to which higher costs pass through to retail prices. This work shows why the effects of trade policy depend on price adjustment at each stage of production and distribution.

Future research. My ongoing work on trade policy brings tariff-rate quotas into quantitative trade models. These quotas apply a low tariff up to a specified import volume and a higher tariff above it, yet most models approximate them with a single tariff. I will study how this policy design affects adjustment to trade shocks and the distribution of gains. I also plan to use the framework of my job market paper to examine how trading rules might be reformed for a changing climate.

On technology regulation, Sandro Steinbach and I are studying why countries reach different approval decisions on similar agricultural technologies, using machine learning and regression analysis of three decades of approval records. On infrastructure, I am extending the Mississippi work to the upper river system to evaluate which maintenance and upgrading investments generate the largest economic benefits. Across these projects, I aim to connect the consequences of policy choices to their design and origins, and to identify ways to support economic adjustment.